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Construction of a forward rate model for the Moroccan interbank market

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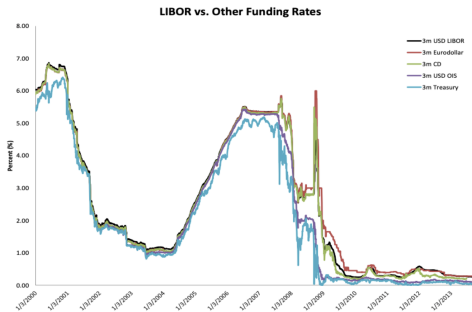
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Motivation

- A topical research theme for Central Bank of Morocco (BAM).
- This research is global in scope.
- Strategic issues for the financial stability of markets.
- Research question: "In the context of the transition from global benchmarks, how can we develop a calculation methodology for these new future maturity risk-free rates that is feasible in the Moroccan context?"

Context of the problem

- The LIBOR rate is considered the "world's most important number" in finance [7].
- Conglomerate of different banks artificially manipulated the rate between 2005 - 2010 [19].



Methodology

- Construction of this forward curve based on data provided by BAM [1].
- Benchmark analysis of work carried out to date by various central banks.
- Consideration of Moroccan market characteristics to select the best approach.
- Application of the compound rate methodology.

Results

- The yield curve fluctuates in a narrow band ranging from 1.420% for a one-week maturity to 1.736% for a projected one-year maturity.
- Like the results of the American team [9], the rate obtained is also not very volatile in the first 6 months modeled.

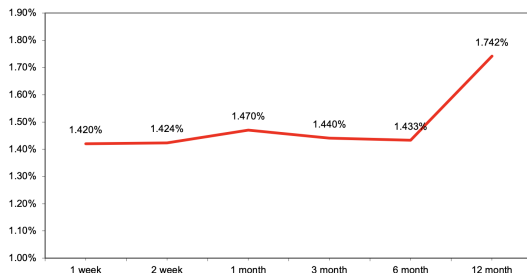


Figure: Computation by the author.

Implication

- Weak financial literature covering this issue, based mainly on the research work of American teams [9].
- We obtain results that are largely in line with the position of scientific research in this respect.
- The extent of MONIA's daily volatility can vary over time and depends on a number of factors, despite which the use of an average overnight rate makes it possible to smooth out almost all these types of volatility.

Limits

- Construction of the forward yield curve using alternative reference rates has not seen the light of day at no central bank until now.
- Paradigm shift in methodology: from a declarative-based rate to an effective-based rate.
- Moroccan market limited in terms of maturity.
- Limiting assumption: the rate is difficult to model for maturities of more than one year. The 12-month rate explodes despite its risk-free nature.
- Impact of Covid-19 on results.
- Research work at a germinal stage.

Conclusion

- The results can be taken as approximation values for forward maturities.
- However, it's worth mentioning that the compounding approach does have its hurdles to overcome.
- This method is the most popular with market operators, as it is easy to understand in terms of theoretical underpinnings, transparent and has a rather stable behavior over time.
- Option to publish the first LIBOR rates in parallel with the new risk-free rates, to smooth the transition and enable agents to prepare themselves better.

References I



Bank Al-Maghrib (BAM), (2020). *Methodology for the Determination of the MONIA (Moroccan Overnight Index Average) Monetary Reference Rate*, Rabat, Morocco: Bank Al-Maghrib (BAM).



Bank Al-Maghrib, (n.d.). MONIA (Moroccan Overnight Index Average). [Online] Available at: [URL needed]



Bank of England (BoE), (2017). SONIA interest rate benchmark. [Online] Available at: <https://www.bankofengland.co.uk/markets/sonia-benchmark> [Accessed 22 March 2020].



Bank of Japan (BoJ), (2016). Public Consultation on Identification and Use of a Japanese Yen Risk-Free Rate. Bank of Japan (BoJ) Publications, 5. [Online] Available at: <https://www.boj.or.jp/en/paym/market/sg/rfr1603c.pdf>.



Bloomberg, (2021). A \$27 Trillion Challenge Looms as Yen Libor Shift Nears. [Online] Available at: [URL needed]

References II

-  Bos, K., (2019). The LIBOR rate transition: On the implementation of transition approaches from Interbank Offered Rates to Risk Free Rates and the corresponding value impact, Twente: University of Twente, Financial Engineering & Management.
-  Hou, D., (2014). LIBOR: Origins, Economics, Crisis, Scandal, and Reform. Report No. 667, 5, New York, USA: FRB of New York Staff.
-  Drake, F. J. & Polonchek, P., (2009). *Finance: Capital Markets, Financial Management and Investment Management*. John Wiley & Sons.
-  Heitfield, E. & Park, Y. H., (2019). "Inferring Term Rates from SOFR Futures Prices," Finance and Economics Discussion Series Divisions of Research & Statistics and Monetary Affairs Federal Reserve Board, Washington, D.C.
-  European Central Bank (ECB), (2017). The euro short-term rate (€STR) methodology and policies. *European Central Bank (ECB) Publications*, 5. Available at: https://www.ecb.europa.eu/paym/initiatives/interest_rate_benchmarks/shared/pdf.

References III



European Central Bank, (n.d.). European Central Bank Statistical Data Warehouse. [Online] Available at: [URL needed].



European Money Markets Institute (EMMI), (2019). Blueprint for the Hybrid Methodology for the Determination of EURIBOR. *European Money Markets Institute Publications*, 4.



EY, (2018). "End of an IBOR era," vol. XXXVII, no. 173, p. 7.



Mishkin, F. M., (2013). What Is Finance? In: *Money, Banking, and Financial Markets 10th Edition (French) (10^{ème} Édition)*, p. 5. Paris, France: Pearson.



Goldman Sachs Markets Research, (2019). LIBOR Transition. [Online] Available at: [URL needed].



ICE Benchmark Administration, (2014). ICE LIBOR METHODOLOGY. *Publication*, 5. Available at:
https://www.theice.com/publicdocs/ICE_LIBOR_Methodology.pdf.



IOSCO, (n.d.). International reference rate principles.

References IV



New York Federal Reserve (FED), (2020). Statement Regarding Publication of SOFR Averages and a SOFR Index. *New York Federal Reserve (FED) Publication*, 1.



Abrantes-Metz, R. M., (2012). "Why and How Should the Libor Be Reformed?" New York University - Leonard N. Stern School of Business - Department of Economics: Global Economics Group, LLC.



Senat, (2009). Economic Survey n° 4 - November 5, 2009 - Monetary policy. Objectives, Methods, and Emerging Issues.



Solnik, C.-D. & B., (2001). "On the Term Structure of Default Premia in the Swap and Libor Market," *The Journal of Finance*, 56, 3 (June), 1095-1115.



Sundarsan, S., (2016). Dynamics of the expectation and risk premium in the OIS term structure. Columbia University - Kelley University at Indiana University (March 1, 2016).



Sushko, V. & Schrimpf, A., (2019). BIS Quarterly Review: Beyond LIBOR: a primer on the new reference rates. Bank of International Settlement Research (BIS).

References V



Swiss Exchange (SIX), (2020). Swiss Index Methodology Rulebook Governing the Swiss Reference Rates. *Swiss Exchange (SIX) Publication*, 5. Available at: https://www.six-group.com/exchanges/downloads/indexinfo/online/swiss_reference_rates/swiss_refe.



Zestos, G. K., (2015). *The Global Financial Crisis (1st ed.)*. New York, USA: Routledge.